

计算机与程序设计基础 (C)

实验课堂

第一讲 运行环境与运行方法

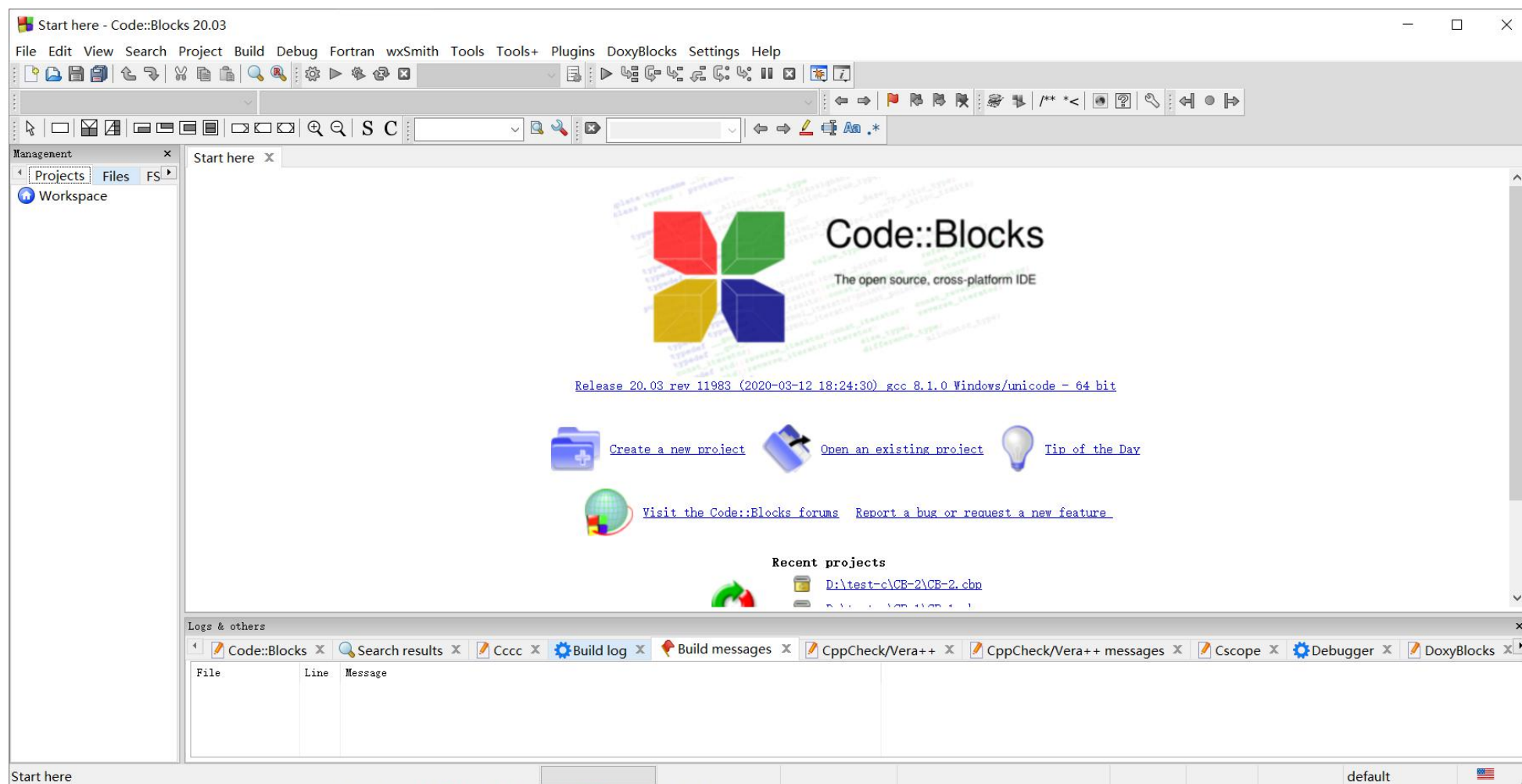


简介

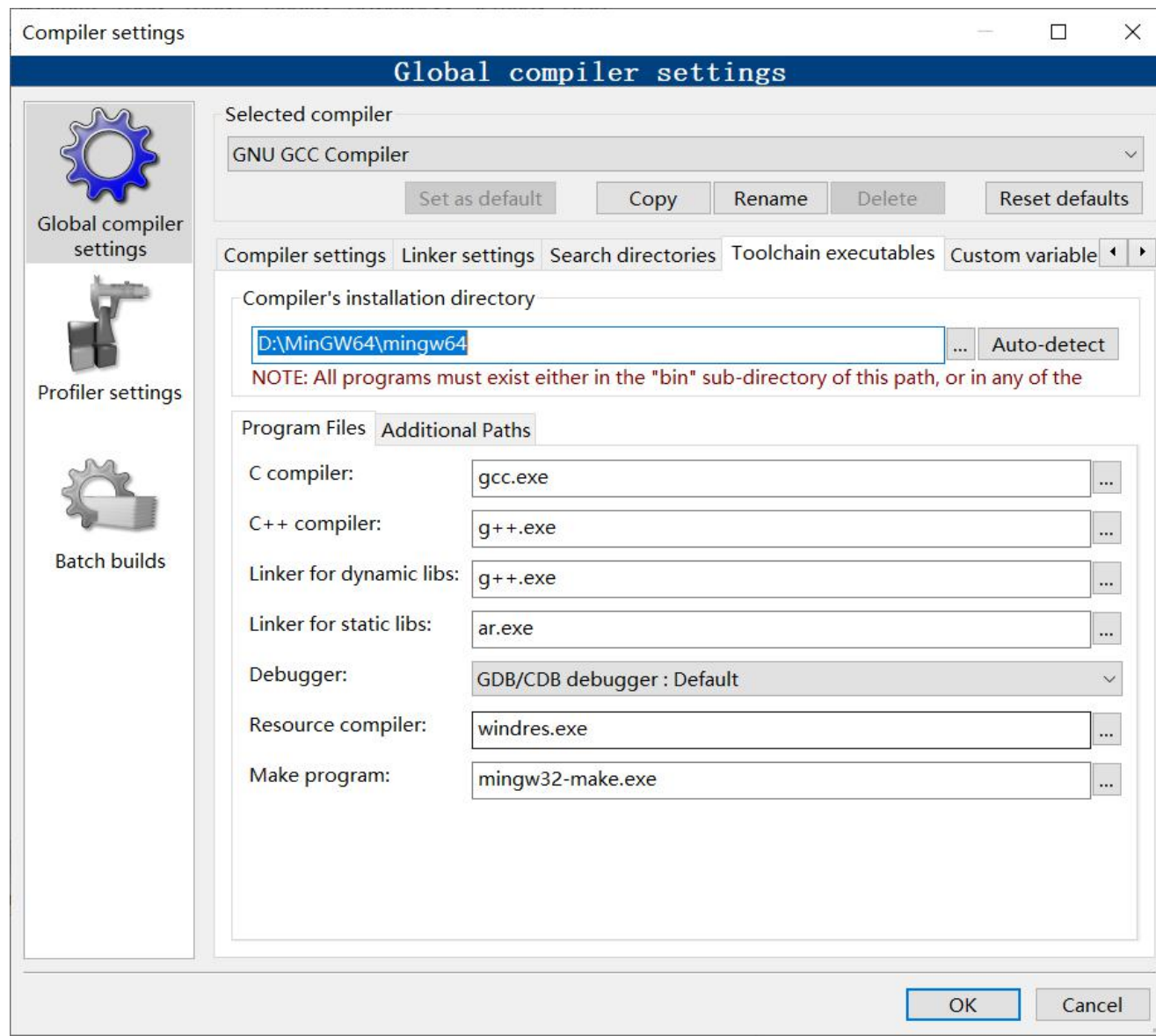
- 实验内容
 - Code::Blocks的安装与使用
 - 初学者常见错误总结
 - 实验任务
- 实验目的
 - 了解C语言集成开发环境Code::Blocks的使用方法
 - 了解如何编辑、编译、链接和运行一个C语言程序
 - 掌握基本的C语言程序调试技能



CodeBlocks的下载、安装及配置



CodeBlocks的下载、安装及配置



CodeBlocks的使用

- 创建工程



[Release 20.03 rev 11983 \(2020-03-12 18:24:30\) gcc 8.1.0 Windows/unicode - 64 bit](#)



[Create a new project](#)



[Open an existing project](#)

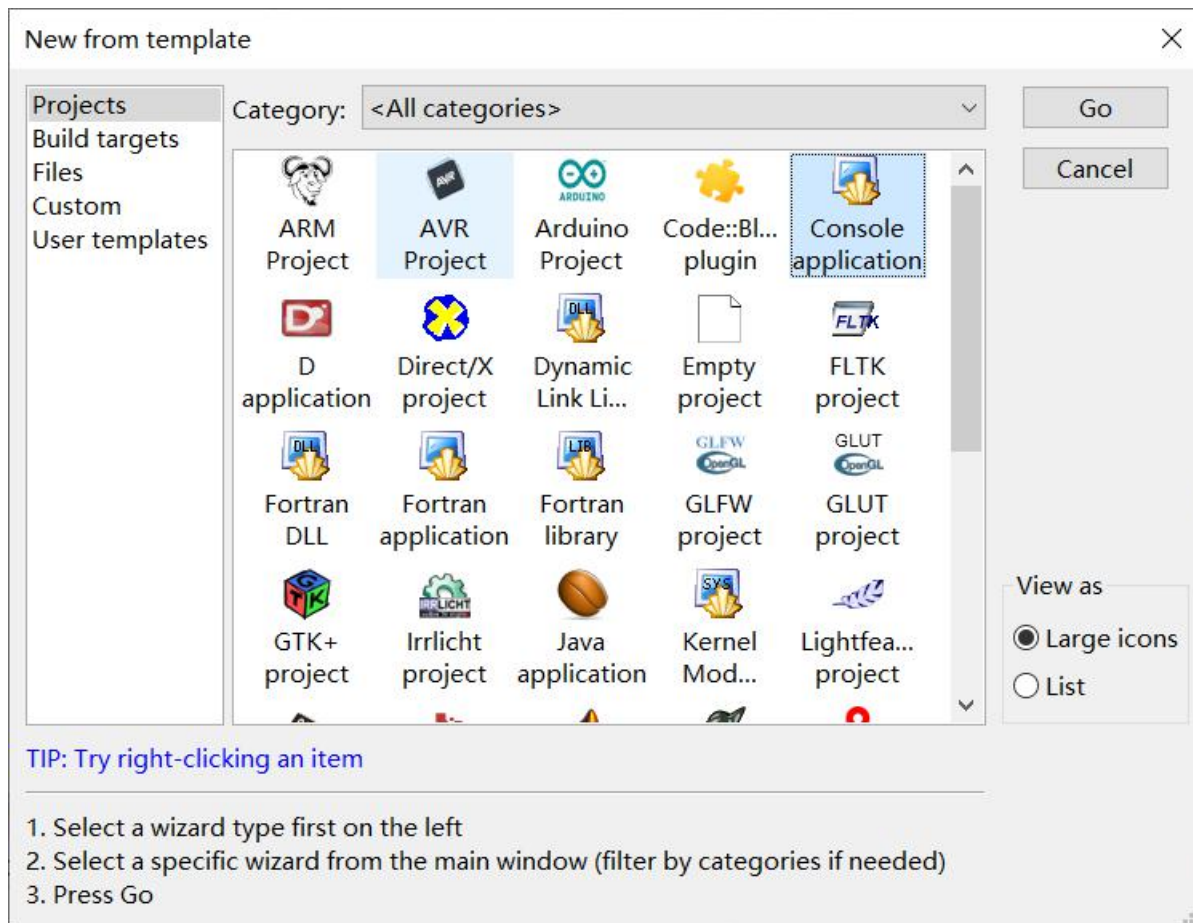


[Tip of the Day](#)



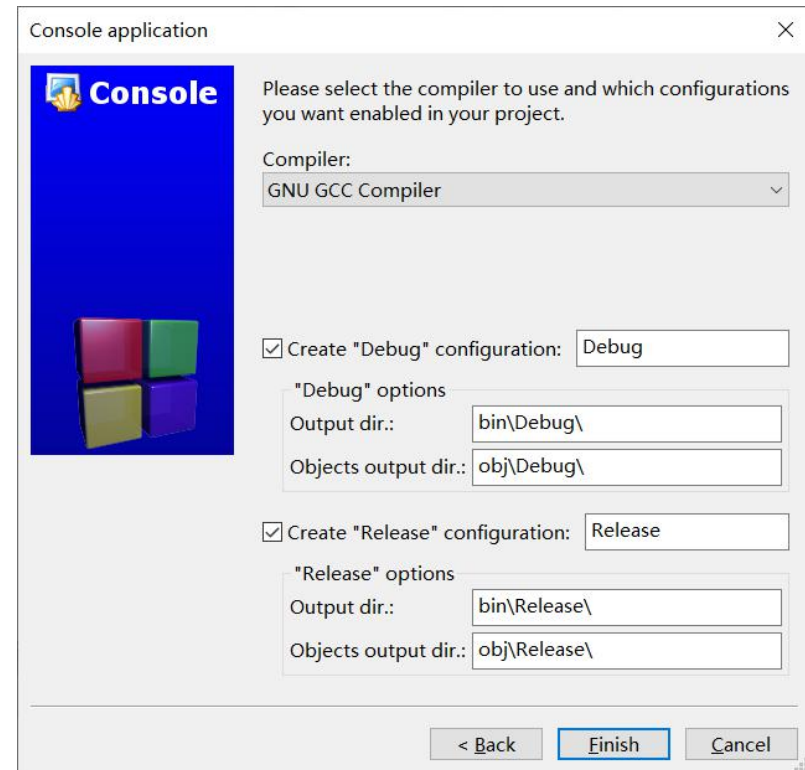
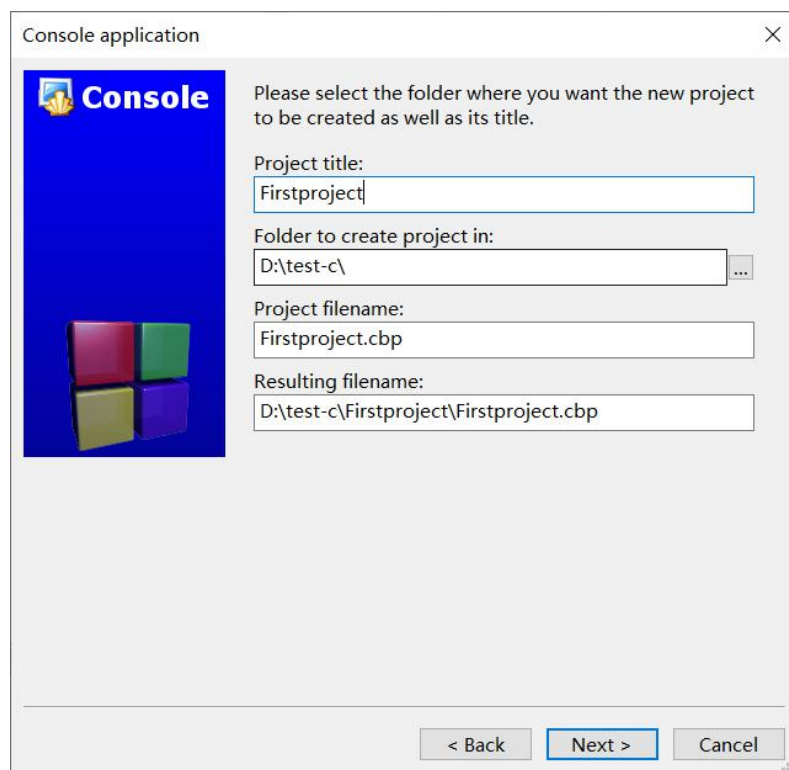
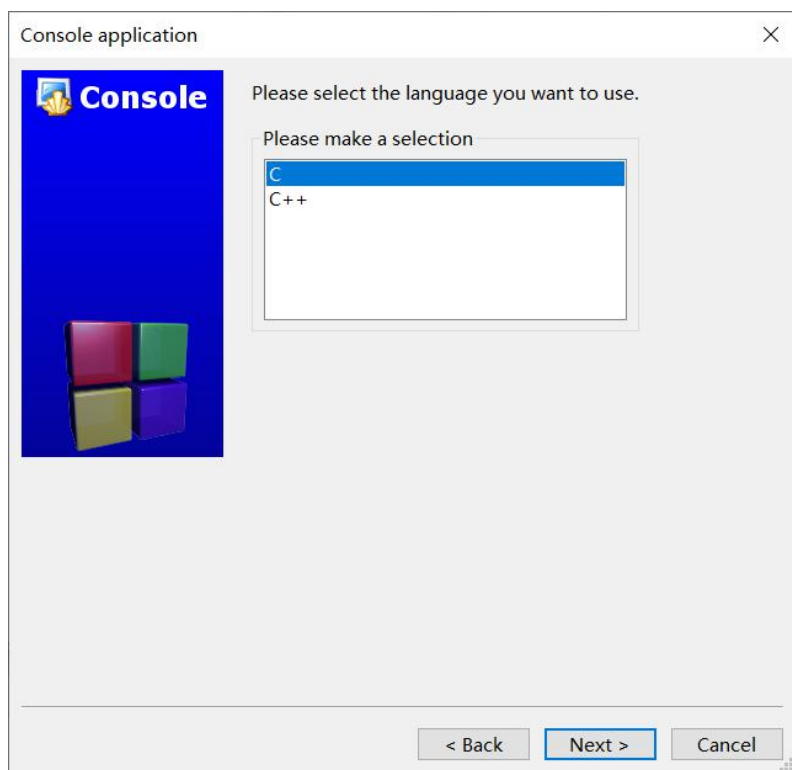
[Visit the Code::Blocks forums](#)

[Report a bug or request a new feature](#)



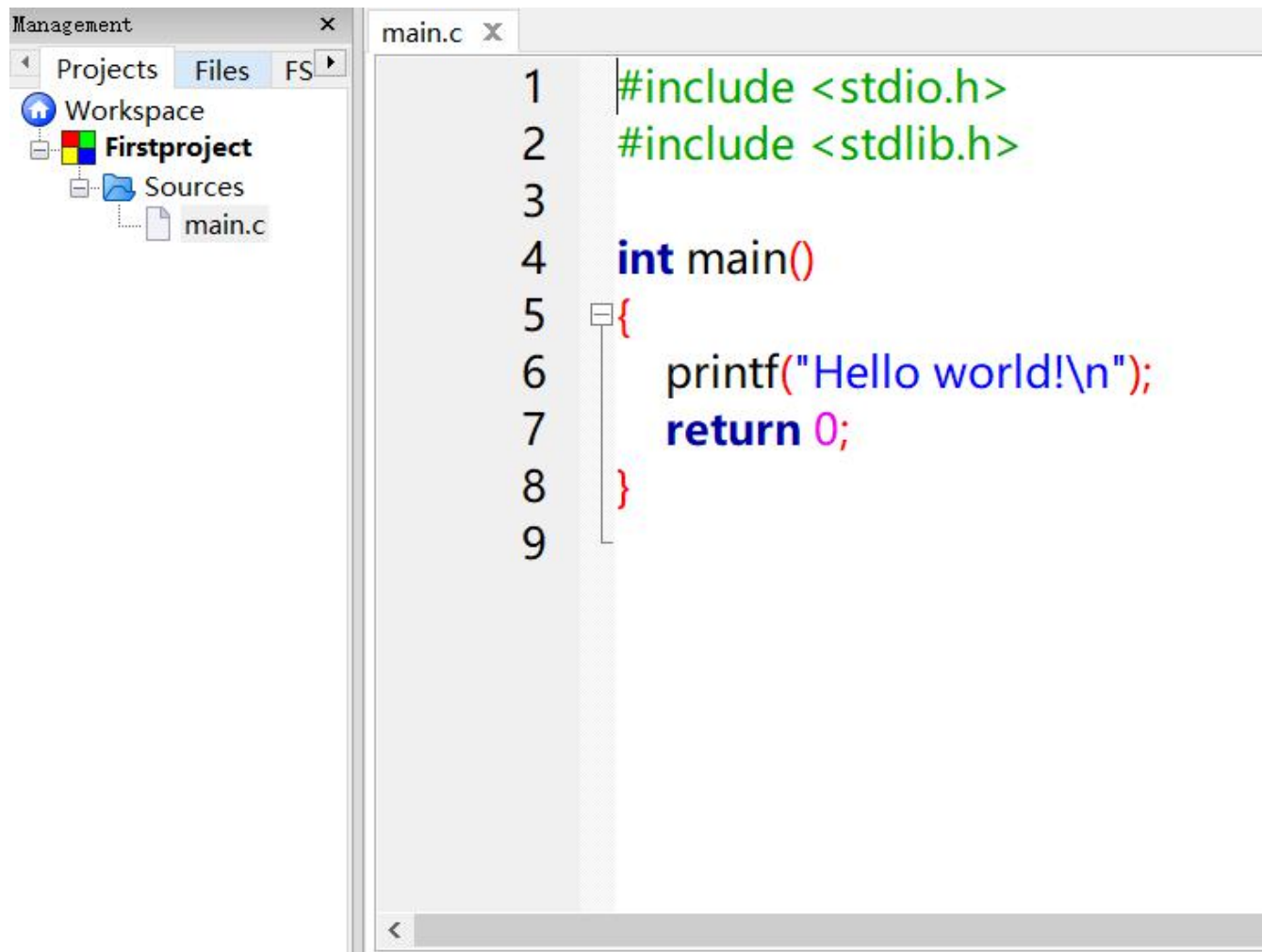
CodeBlocks的使用

- 创建工程



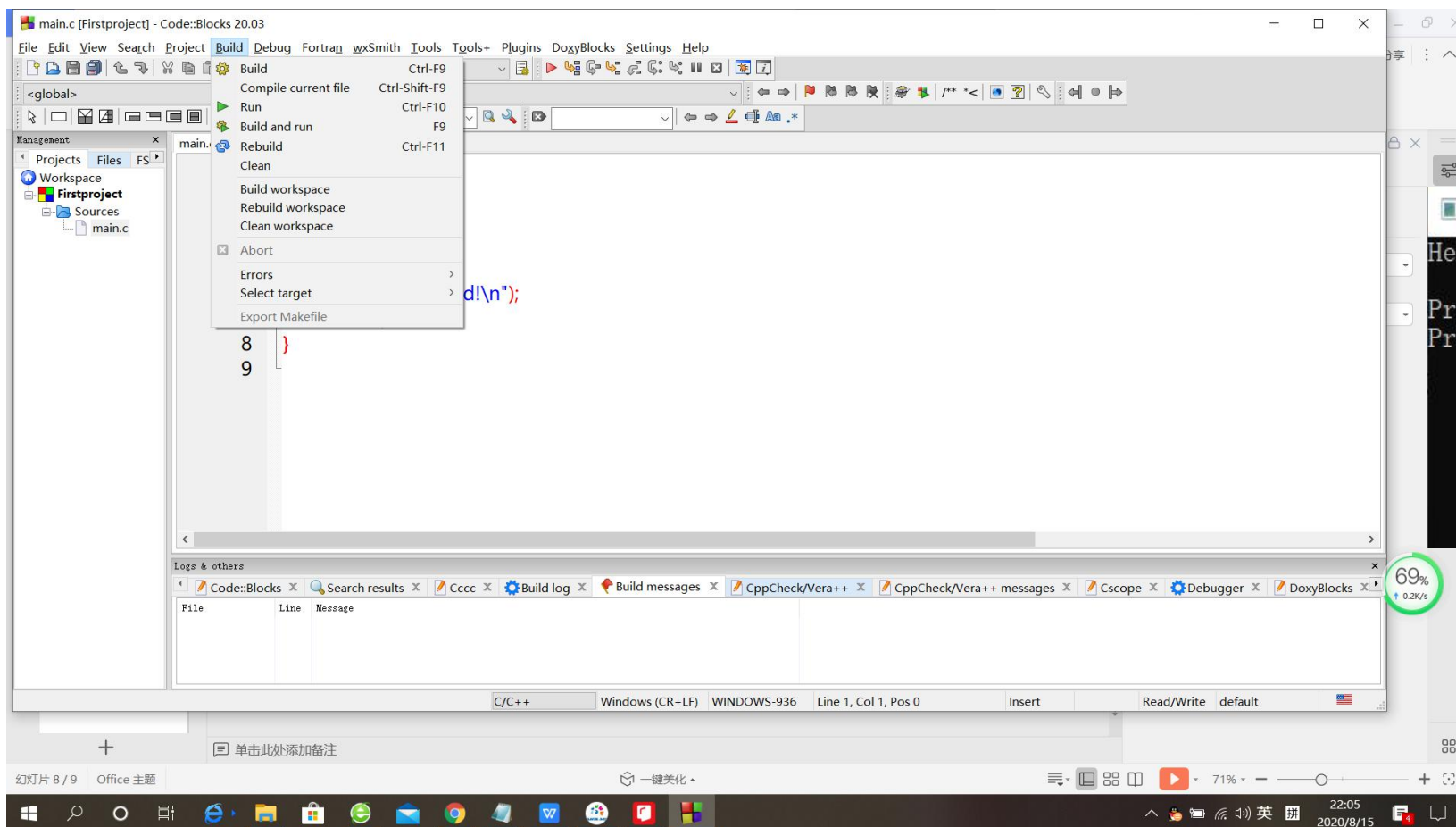
CodeBlocks的使用

- 编写源文件



CodeBlocks的使用

- 编译、链接和运行



D:\test-c\Firstproject\bin\Debug\Firstproject.exe

Hello world!

Process returned 0 (0x0) execution time : 2.567 s
Press any key to continue.

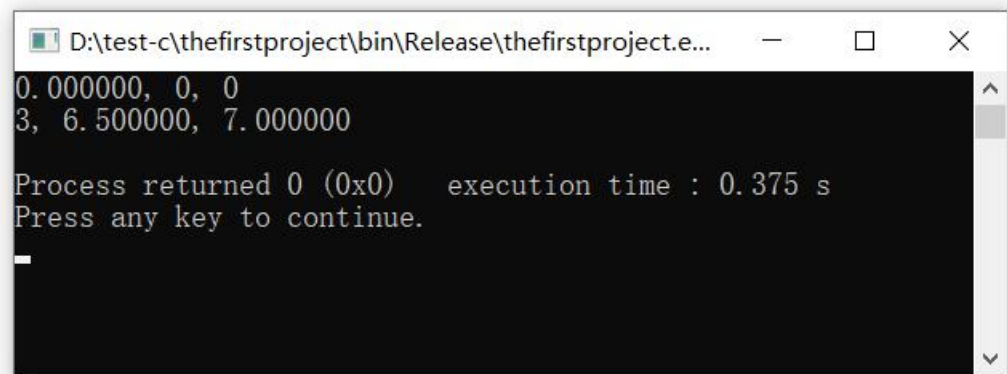
常见错误1：忘了定义变量

```
1  #include <stdio.h>
2
3  int main()
4  {
5      x=3;
6      y=6;
7      printf("%d\n", x+y);
8      return 0;
9  }
10
```

File	Line	Message
		=== Build: Release in thefirstproject (compiler: GNU GCC Compiler) ===
D:\test-c\the...		In function 'main':
D:\test-c\the...	5	error: 'x' undeclared (first use in this function)
D:\test-c\the...	5	note: each undeclared identifier is reported only once for each function it appears in
D:\test-c\the...	6	error: 'y' undeclared (first use in this function)
		=== Build failed: 2 error(s), 0 warning(s) (0 minute(s), 0 second(s)) ===

常见错误2：类型不一致

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int x=3;
6      float y=6.5f;
7      double z=7.0;
8      printf("%f, %d, %d\n", x, y, z);
9      printf("%d, %f, %f\n", x, y, z);
10     return 0;
11 }
12
```



```
D:\test-c\thefirstproject\bin\Release\thefirstproject.e...
0.000000, 0, 0
3, 6.500000, 7.000000

Process returned 0 (0x0)   execution time : 0.375 s
Press any key to continue.
```

常见错误3：未考虑取值范围

```
1  #include <stdio.h>
2  #include <limits.h>
3  int main()
4  {
5      short num1;
6      int num2;
7      num1 = num2 = 89101;
8      printf("num1=%d, num2=%d\n", num1, num2);
9      printf("short's max is %d\n", SHRT_MAX);
10     return 0;
11 }
12
```

D:\test-c\2021c课程\t1\bin\Debug\t1.exe

num1=23565, num2=89101
short's max is 32767

Process returned 0 (0x0) execution time : 0.453 s
Press any key to continue.

常见错误4：乱加分号

```
1  #include <stdio.h>;
2
3  int main()
4  {
5      if(0);
6          printf("Hello world!");
7      return 0;
8  }
```

```
D:\test-c\thefirstproject\main.c:1:19: warning: extra tokens at end of #include directive
```

```
#include <stdio.h>;
      ^
```

```
D:\test-c\thefirstproject\main.c: In function 'main':
```

```
D:\test-c\thefirstproject\main.c:5:5: warning: this 'if' clause does not guard... [-Wmisleading-indentation]
```

```
    if(0);
    ^^
```

```
D:\test-c\thefirstproject\main.c:6:9: note: ...this statement, but the latter is misleadingly indented as if it were guarded by the 'if'
```

```
        printf("Hello world!");
        ~~~~~
```

常见错误5：大小写不分

```
1  #include <stdio.h>;
2
3  int main()
4  {
5      int a, b, c;
6      A=B=C=100;
7      printf("%d, %d, %d\n", A, B, C);
8      return 0;
9  }
```

```
D:\test-c\the... 1 warning: extra tokens at end of #include directive
D:\test-c\the... In function 'main':
D:\test-c\the... 6 error: 'A' undeclared (first use in this function)
D:\test-c\the... 6 note: each undeclared identifier is reported only once for each function it appears in
D:\test-c\the... 6 error: 'B' undeclared (first use in this function)
D:\test-c\the... 6 error: 'C' undeclared (first use in this function)
D:\test-c\the... 5 warning: unused variable 'c' [-Wunused-variable]
D:\test-c\the... 5 warning: unused variable 'b' [-Wunused-variable]
D:\test-c\the... 5 warning: unused variable 'a' [-Wunused-variable]
=== Build failed: 3 error(s), 4 warning(s) (0 minute(s), 0 second(s)) ===
```


常见错误6：使用中文标点符号

```
1  #include <stdio.h>;
2
3  int main()
4  {
5      int a, b, c;
6      a=b=c=100;
7      printf("%d, %d, %d\n", a, b, c);
8      return 0;
9  }
```

D:\test-c\the...	5	error: stray '\243' in program
D:\test-c\the...	5	error: stray '\254' in program
D:\test-c\the...	5	error: expected '=', ',', ';', 'asm' or '__attribute__' before 'b'
D:\test-c\the...	5	error: 'b' undeclared (first use in this function)
D:\test-c\the...	5	note: each undeclared identifier is reported only once for each function it appears in
D:\test-c\the...	5	error: 'c' undeclared (first use in this function)
D:\test-c\the...	5	warning: left-hand operand of comma expression has no effect [-Wunused-value]
D:\test-c\the...	6	error: 'a' undeclared (first use in this function)
D:\test-c\the...	6	error: stray '\243' in program
D:\test-c\the...	6	error: stray '\273' in program
D:\test-c\the...	6	error: expected ';' before 'printf'
=== Build failed: 9 error(s), 2 warning(s) (0 minute(s), 1 second(s)) ===		

任务一：打印信息

```
1  #include <stdio.h>
2
3
4  int main(void)
5  {
6      int num;
7      num = 1;
8      printf ("My favorite number is %d\n", num);
9      return 0;
10 }
```


任务二*：接受输入参数

```
1  #include <stdio.h>
2  //Project_>set program's arguments...
3
4  int main(int argc, char *argv[])
5  {
6      int i;
7      printf("%d\n", argc);
8      for(i=0; i<argc; i++)
9          printf("%d:  %s\n", i, argv[i]);
10     return 0;
11 }
```

任务三：输入输出

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int a, b;
6      double c;
7      float d;
8      printf("please input a, b, c and d:\n");
9      scanf ("%d %d %lf, %f", &a, &b, &c, &d);
10     printf ("a=%d, b=%d, c=%f, d=%f\n", a, b, c, d);
11 }
```

这里有个陷阱!

```
please input a, b, c and d:
1 2 3.0 4
a=1, b=2, c=3.000000, d=0.000000
```

```
please input a, b, c and d:
1 2 3, 4
a=1, b=2, c=3.000000, d=4.000000
```

任务四：体验函数

```
1  #include <stdio.h>
2
3  int max (int x, int y)
4  {
5      int z;
6      if (x>y)
7          z = x;
8      else
9          z = y;
10     return z;
11 }
12
13
14 int main()
15 {
16     int a, b, c;
17     printf ("Input a and b\n");
18     scanf ("%d, %d", &a, &b);
19     printf ("a=%d, b=%d\n", a, b);
20     c = max (a, b);
21     printf ("max=%d\n", c);
22 }
```

```
Input a and b
1, 100
a=1, b=0
max=1
```



```
Input a and b
1 100
a=1, b=0
max=1
```



```
Input a and b
1,100
a=1, b=100
max=100
```

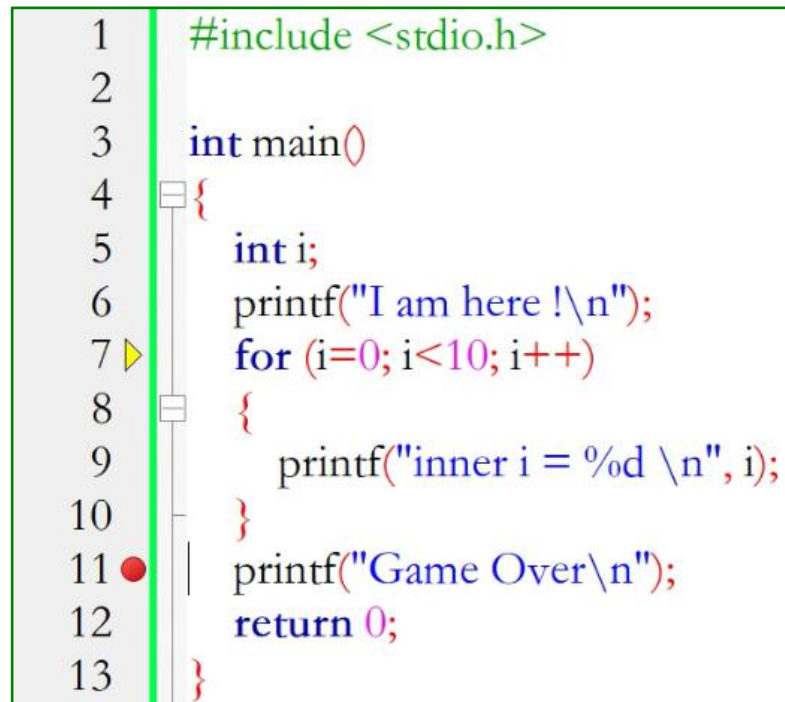


任务五*: 错误排查

```
1  #include <stdio.h>
2  //many errors. Find them!
3  int main()
4  {
5      int i=1;
6      printf("old i=%d,\n", i);
7      i=add(i);
8      printf("new i=%f,\n", i);
9  }
10
11 void add10(int x)
12 {
13     x+=10
14     printf("in add10 function\n");
15     return x;
16 }
```

任务六*：调试技巧

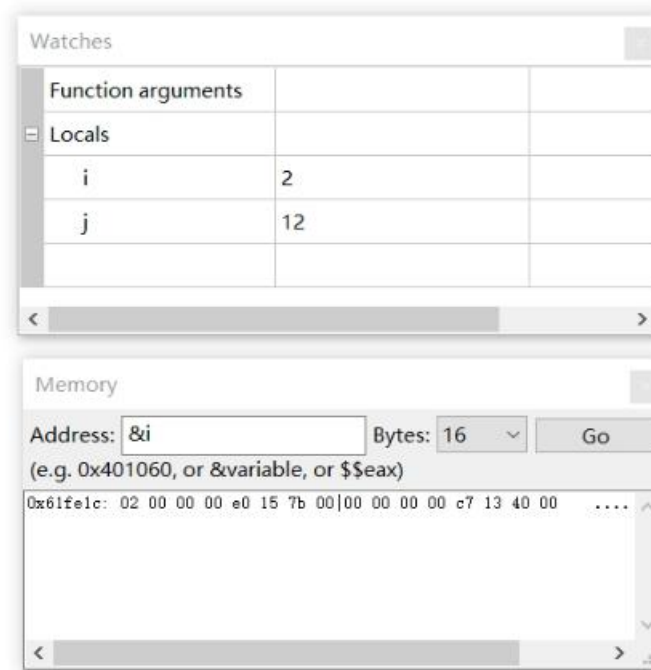
```
1  #include <stdio.h>
2
3  int main()
4  {
5      int i;
6      printf("I am here !\n");
7      for (i=0; i<10; i++)
8      {
9          printf("inner i = %d \n", i);
10     }
11     printf("Game Over\n");
12     return 0;
13 }
```



```
1  #include <stdio.h>
2
3  int main()
4  {
5      int i;
6      printf("I am here !\n");
7  ▶  for (i=0; i<10; i++)
8      {
9          printf("inner i = %d \n", i);
10     }
11  ●  printf("Game Over\n");
12     return 0;
13 }
```

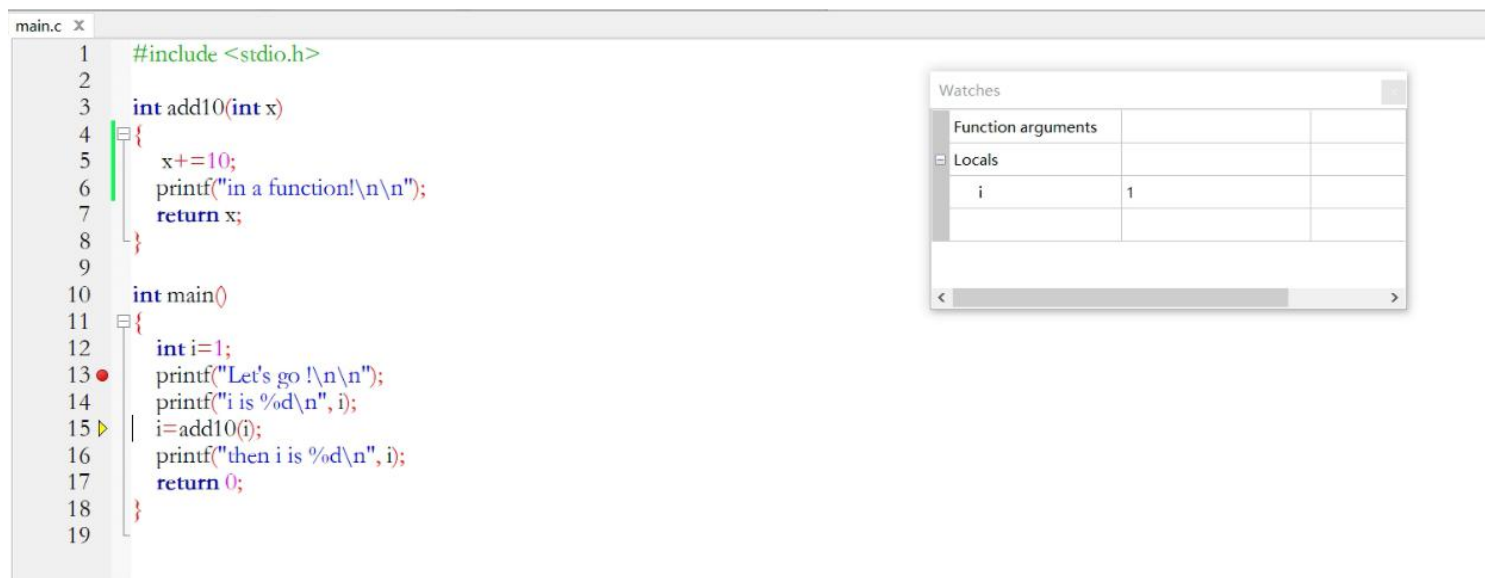

任务六*：调试技巧

```
1  #include <stdio.h>
2  int main()
3  {
4      int i=0;
5      double j=10.0;
6      printf("Let's go !\n\n");
7      printf("Size of double= %d \n", sizeof(j));
8      printf("Address of i=%p\n", &i);
9      printf("Address of j=%p\n", &j);
10     for (i=0; i<10; i++)
11     {
12         printf("i = %d, j=%f \n", i, j);
13         j++;
14     }
15     printf("\n Game Over\n");
16     return 0;
17 }
```



任务六*: 调试技巧

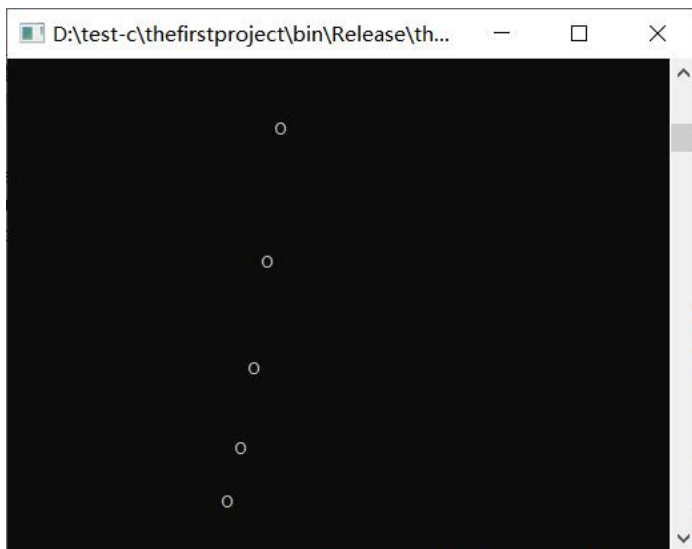
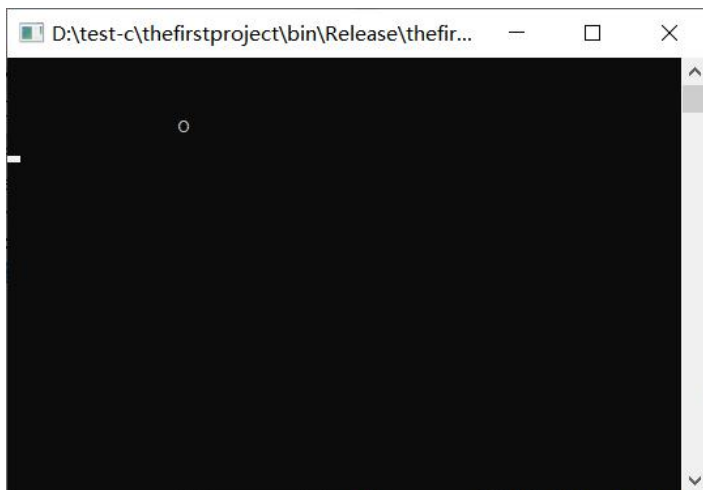
```
1  #include <stdio.h>
2
3  int add10(int x)
4  {
5      x+=10;
6      printf("in a function!\n\n");
7      return x;
8  }
9
10 int main()
11 {
12     int i=1;
13     printf("Let's go !\n\n");
14     printf("i is %d\n", i);
15     i=add10(i);
16     printf("then i is %d\n", i);
17     return 0;
18 }
```



The screenshot shows a code editor with the same C code as the previous block. The code is color-coded and includes line numbers. A debugger window is open on the right side of the editor, showing the 'Watches' panel. The 'Watches' panel has a table with the following content:

Watches	
Function arguments	
Locals	
i	1

任务七*: 小游戏之反弹小球



```
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <windows.h>
4  int main()
5  {
6      int i,j;
7      int x = 0;
8      int y = 5;
9
10     int velocity_x = 1;
11     int velocity_y = 1;
12     int left = 0;
13     int right = 20;
14     int top = 0;
15     int bottom = 10;
16
17     while (1)
18     {
19         x = x + velocity_x;
20         y = y + velocity_y;
21
22         system("cls"); // clean screen
23         // print some spaces
24         for(i=0;i<x;i++)
25             printf("\n");
26         for (j=0;j<y;j++)
27             printf(" ");
28         printf("o"); // print the ball
29         printf("\n");
30         Sleep(50); // waiting
31
32         if ((x==top)||(x==bottom))
33             velocity_x = -velocity_x;
34         if ((y==left)||(y==right))
35             velocity_y = -velocity_y;
36     }
37     return 0;
38 }
```